

SET: A

Name	
Roll	
Signature	
Lecture Hall Number	

1) Time duration in minutes (i) SIXTY (60) (non DAP) / EIGHTY (80) (DAP) students.

2) Closed Notes/Book/Slide examination.

3) No mobiles or any other device allowed.

4) Only calculator and pen/pencil allowed

5) Exchange of calculator, pen/pencil between students not allowed.

6) Total number of questions is **TWENTY (20)** and the questions are

- ◇ Exact answer (Quantitative).
- ◇ MCQ (FIVE (05) options with ONE (01) CORRECT answer (Qualitative).

7) Marks: CORRECT is +2/Plus TWO; WRONG is -1/Minus ONE; No ANSWER is 0

8) Total marks **FORTY (20×2=40)**.

FOR OFFICIAL USE

	CORRECT	WRONG	NOT ATTEMPTED	MARKS
QUANTITATIVE				
Q 01 to # 15				
MCQ				
Q # 16 to # 20				

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QUESTION # 01 to # 15: WRITE THE EXACT INTEGER VALUED ANSWER	
01	
02	
03	
04	
05	
06	
07	
08	
09	
10	
11	
12	
13	
14	
15	

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QUESTION # 16 to # 20: CIRCLE THE BEST OPTION (QUALITATIVE)					
16	Degenerate	Unique optimal	and	Unbounded	Alternative or Multiple Optimal Infeasible
17	Degenerate	Unique optimal	and	Unbounded	Alternative or Multiple Optimal Infeasible
18	Degenerate	Unique optimal	and	Unbounded	Alternative or Multiple Optimal Infeasible
19	Degenerate	Unique optimal	and	Unbounded	Alternative or Multiple Optimal Infeasible
20	Degenerate	Unique optimal	and	Unbounded	Alternative or Multiple Optimal Infeasible

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Q # 01

From the calculation it was found that break-even point (BEP) was 9,000 units. It is given that fixed cost (FC) is Rs. 36,000/-, revenue per unit (r) is Rs. 9/-, then the variable cost per unit (v) (in Rs.) is?

Q # 02

Demand is 260 units per 10 minutes while capacity per machine is 90 units per 15 minutes. Find the flow rate (units per hour) considering there are FIVE (05) machines each of the same efficiency and they are used simultaneously, separately and are operating parallel to each other?

Q # 03

You are given demand rate of item (D) as 2,000 units per year; fixed cost (A) for each replenishment is Rs. 2,000/-; variable cost (v) as Rs. 4/-per unit and carrying cost (r) as 2 per year. Find the time between orders (in months) when you order EOQ number of units each time.

Q # 04

For $\min: f(\mathbf{x}) : -5x_1 - 4x_2 - 6x_3$

$$x_1 - x_2 + x_3 \leq 20$$

$$3x_1 + 2x_2 + 4x_3 \leq 40$$

$$3x_1 + 2x_2 + 0x_3 \leq 30$$

$$x_1, x_2, x_3 \geq 0$$

At the optimal value (minimum of $f(\mathbf{x}^*) = f(x_1^*, x_2^*, x_3^*)$) the corresponding value for x_2^* is?

Q # 05

For $\max: f(\mathbf{x}) : 5x_1 + 2x_2$

$$x_1 + x_2 \leq 6$$

$$x_1 - x_2 \leq 0$$

$$x_1, x_2 \geq 0$$

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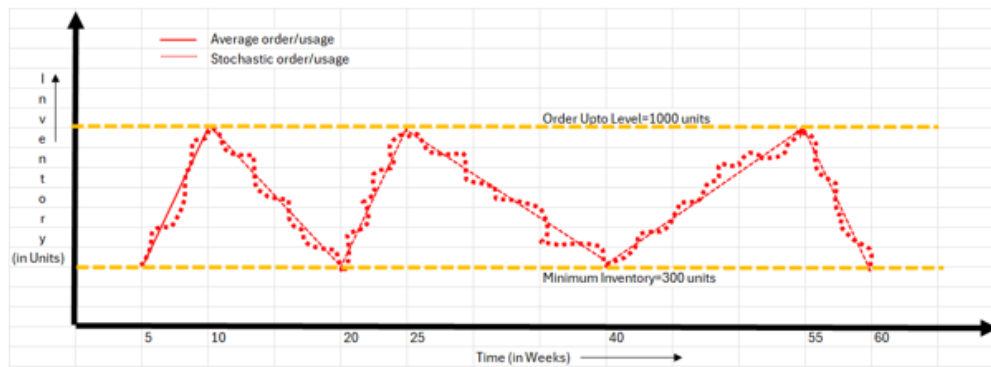
The optimal value (maximum in this case) for $f(\mathbf{x}^*) = f(x_1^*, x_2^*)$ is?

Q # 06

In PERT Network a job/activity has the following completion times which are (i) Optimistic = 12 days, (ii) Most Likely = 3 weeks and (iii) Pessimistic = 6 weeks. Given this information the expected completion time for the same job/activity (in days) is?

Q # 07

For the following graph (Inventory Level (**depicted along Y-axis**) versus Time (**depicted along X-axis**)), what is the average inventory level (in units/week) between the starting of 5th week to beginning of 60th week?



Q # 08

Mr. Gajendra Pratap has a kachori stall which is very popular in the city of Benaras. It takes about 1.0 minutes to take an order and collect the payment, while the time taken to prepare one order (i.e., 1 plate which serves two (02) kachoris and daal) is 1.5 minutes per order. Selling price per plate is Rs. 40/- while material costs per plate is Rs. 10/- and there are no fixed costs. The shop operates from 0700 hrs (IST) to 1200 hrs (IST) the day it is open. Considering the demand Mr. Pratap wishes to hire ONE (01) extra person that will reduce the per order (a plate which serves two (02) kachoris and daal) time from 1.5 minutes/90 seconds to 80 seconds. The person hired will have to be paid wage at the rate of Rs. 500/- for the total FIVE (05) days in the week the shop is open. From a purely financial perspective, what **profit** Mr. Pratap makes per week (week implies Wednesday to Monday, i.e., the shop is open SIX (06) days a week) once he hires the helper?

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Q # 09

Consider the example of the three-step airport security. The **FIRST** step, verifying ID and boarding pass, takes 30 seconds per passenger. The **SECOND** step, searching the passenger for metal objects using a metal detector, takes 10 seconds per passenger. The **THIRD** step, running the carry-on baggage through an X-ray machine, takes 60 seconds per passenger. Assume that there are many customers waiting in the process and that **EACH** of the **THREE (03)** employees (who are employed to man and run the three-step airport security system) gets paid Rs. 1500/- per hour. Then what is the cost of direct labor (calculate the value as Rs. per passenger)?

Q # 10

For a production process the value of fixed cost (A) is Rs. 1500/-, Demand (D) is 6000 units per year, variable cost (v) per unit is Rs. 9/- and carrying cost (r) is 2. What is the total cost (in INR) when order quantity is 20% **more** than the **economic order quantity (EOQ)**?

Q # 11

A machine-paced line has **SIX (06)** stations with processing times of 30, 20, 25, 28, 22, and 24 seconds per unit, respectively. The process is **capacity-constrained** and runs on a 30 second/unit cycle time. Given this information what is the total idle time (in seconds) across all **SIX (06)** stations?

Q # 12

Assume Demand is 20 units per 5 minutes while Capacity per machine is 30 units per 10 minutes, and this is **ONE (01)** machine only. Find the flow rate/throughput (in units per hour).

Q # 13

SBI offers customized loans. Customers have specific loan request and when they visit the bank are able to obtain a response on an average within 30 minutes. Queries for SBI loan include five activities that must be conducted in the sequence described below.

- **Activity 1:** Queries related to why loan is required by a person and is answered in 4 minutes.
- **Activity 2:** The SBI counter person gathers all the related information from the customer on an average in 5 minutes.
- **Activity 3:** The SBI counter person analyzes the information and checks the papers/documents, etc. of the customer to decide the loan and amount and on an average it takes 7 minutes.
- **Activity 4:** The SBI counter person finally checks and verifies all the papers/documents and on an average it takes 2 minutes.
- **Activity 5:** Finally the customer signs the documents and has the loan money transferred to his/her account and on an average it takes 4 minutes.

The whole process is conducted by three (03) SBI persons in a worker-paced line. The assignment of tasks to SBI personnel is as follows: **W1 does activity 1, W2 does activities 2 and 3, and W3 does activities 4 and 5.** Assume each SBI person is paid the same wages which is Rs. 2000/- per hour. You can assume that there exists **unlimited demand** and that loans are only entering the process at the rate of the bottleneck.

Then what is cost of direct labour (in Rs. per loan)?

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Q # 14

A primary hospital has the **capacity** to see 75 patients per 1 hour. The average **demand** rate is 385 patients per 5 hours. What is the utilization in percentage for the hospital?

Q # 15

Given $P = \text{Rs. } 100$, $r = 10\%$ pa (Simple Interest Rate) and $n = 1$ yr, the amount (in Rs.) after 1 year is?

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Q # 16

For the LPP formulation

$$\text{Max } f(\mathbf{x}) = \text{Max } f(x_1, x_2) = \text{Max: } 3x_1 + 9x_2$$

$$\text{s.t.: } x_1 + 4x_2 \leq 8$$

$$x_1 + 2x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

The solution(s) is/are

Q # 17

For the LPP formulation

$$\text{Max } f(\mathbf{x}) = \text{Max } f(x_1, x_2) = \text{Max: } 3x_1 - x_2$$

$$\text{s.t.: } 2x_1 + x_2 \geq 2$$

$$x_1 + 3x_2 \leq 3$$

$$x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

The solution(s) is/are

Q # 18

For the LPP formulation

$$\text{Max } f(\mathbf{x}) = \text{Max } f(x_1, x_2) = \text{Max: } +5x_1 + x_2$$

$$\text{s.t.: } 2x_1 + 3x_2 \geq 6$$

$$3x_1 + 2x_2 \geq 6$$

$$x_1, x_2 \geq 0$$

The solution(s) is/are

Q # 19

For the LPP formulation

$$\text{Min } x f(\mathbf{x}) = \text{Min } f(x_1, x_2) = \text{Min: } +4x_1 + 2x_2$$

$$\text{s.t.: } 2x_1 + x_2 \geq 10$$

$$x_1 + 2x_2 \geq 8$$

$$x_1, x_2 \geq 0$$

The solutions is/are

Q # 20

For the LPP formulation

$$\text{Max } f(\mathbf{x}) = \text{Max } f(x_1, x_2) = \text{Max: } +5x_1 + 4x_2$$

$$\text{s.t.: } 2x_1 + 3x_2 \leq 6$$

$$5x_1 + 4x_2 \geq 20$$

$$x_1, x_2 \geq 0$$

The solution(s) is/are